Environmental Studies — Week 6

Energy Resources: Solar, Biomass, Hydrogen (with quick recall of Hydro, Wind, Ocean, Geothermal, Law)

Illustrated notes: mechanism first, then diagram, then worked example

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1 Energy resources, net energy and the scale of the sun

Definition

Energy comes from the Greek *en-ergon*, "in work". **Non-renewable** resources (oil, petroleum, diesel, coal, natural gas) are exhaustible; **renewable** resources (solar, geothermal, wind, biomass, hydro, tidal) are replaced as fast as, or faster than, we use them. **Net energy** = useful energy produced − energy needed to develop it.

History in the course notes. Humans began with firewood and animals; Franklin discovered electricity; Faraday built the dynamo in 1831 (mechanical → electrical); Edison invented the light bulb; Tesla the AC motor (electrical → mechanical). A country's energy consumption is usually read as a sign of its development.

Fixing a common misconception

"Energy is used up." Energy is never destroyed, only converted to less useful forms (mostly heat). "Resource" questions are really about *quality*: can we get concentrated, controllable energy out of it for less energy than we put in?

"Renewable means no environmental cost." Renewable describes the *supply*; every system still has manufacturing, land, wildlife or material costs (you will see these in each advantages/disadvantages list).

1.1 Net energy: output must clearly beat input, and the output must be clean

WHY this is true

Define EROI $= E_{\text{out}}/E_{\text{in}}$ and net energy $= E_{\text{out}} - E_{\text{in}}$. Two sources can have the same output but very different net energy.

Worked example

Source A: out = 100, in = 10 ⇒ net = 90, EROI = 10. Source B: out = 100, in = 80 ⇒ net = 20, EROI = 1.25.
B delivers little surplus for society. If its EROI fell below 1 it would be an energy *sink*, not a resource. This is the course's argument against nuclear (large energy for uranium mining and enrichment, fuel-rod fabrication, safe storage of used rods ⇒ low net energy). Coal has higher net energy than nuclear in the notes, but burning it releases CO_2 and other gases, so **high net energy alone is not enough**.

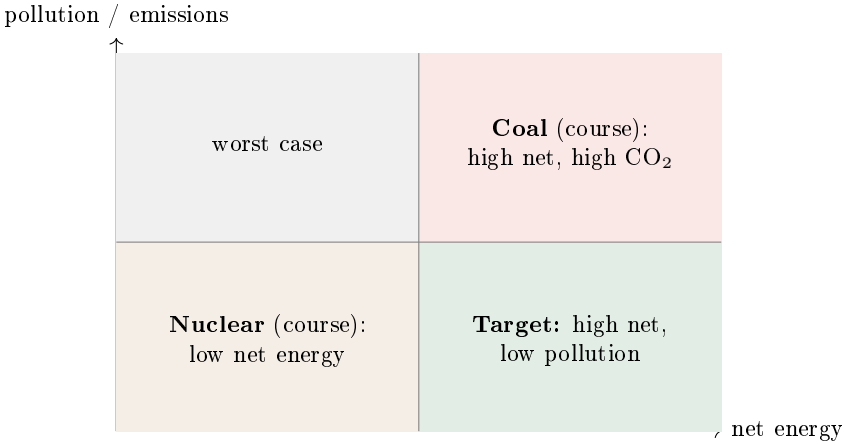


Fig. 1. The two tests from the notes. Positions are schematic and follow the course's reasoning about coal and nuclear.

1.2 Power versus energy, and how big is the sun?

WHY this is true

Power is a rate (watt, W); energy is power × time (watt-hour, Wh; 1 kWh = 3.6 MJ). The sun's power reaching Earth is $S_0 \pi R_E^2$ with $S_0 = 1361$ W/m^2 and $R_E = 6.371 \times 10^6$ m: $1361 \times 1.275 \times 10^{14} = 1.74 \times 10^{17}$ W = 174,000 TW. World primary energy use is about 16 TW (the course chart). Ratio $\approx 10^4$; equivalently, one *year* of world energy use ($16 \times 10^{12} \times 3.16 \times 10^7 = 5.0 \times 10^{20}$ J) equals about $5.0 \times 10^{20} / 1.74 \times 10^{17} \approx 2900$ s, roughly **50 minutes** of intercepted sunlight. (The course chart lists ∼ 23,000 TW for the practically usable solar resource; either way it exceeds demand by over a thousand times.)

Worked example

Capacity factor. Bhadla Solar Park (Rajasthan; 14,000 acres; 2250 MW): 2250 MW × 8760 h = 19.7 TWh if it ran at full power all year. Solar produces nothing at night and less in cloud, so with a capacity factor of ∼ 25% (illustrative): $19.7 \times 0.25 \approx 4.9$ TWh/yr. Land use: $14{,}000 \times 4047$ m^2 $= 5.67 \times 10^7$ m^2, so 2250 MW/56.7 km^2 ≈ 40 MW/km^2.

2 Direct and indirect solar energy

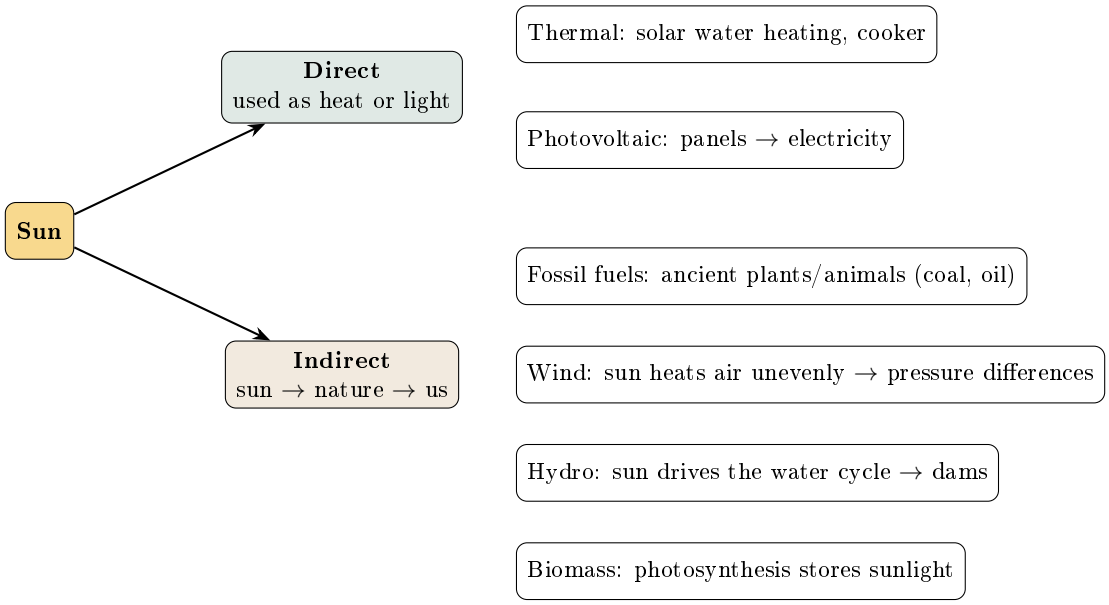


Fig. 2. Everything except geothermal (Earth's internal heat), tidal (Moon's gravity) and nuclear traces back to the sun.

3 Passive solar: no pumps, no moving parts

Definition

A **passive** system collects and uses sunlight without mechanical devices. Course examples: evaporating seawater for salt, drying food and clothes, the **solar box cooker**, and **daylighting** (south-facing windows).

WHY this is true

A dark surface absorbs sunlight and warms. In a box cooker the inner chamber is painted black, the box is insulated, and a glass cover lets visible light in. Hot surfaces re-emit *infrared*, which glass blocks, and the insulation and cover stop warm air escaping. This is the same trap as Week 5's greenhouse effect on a small scale. Temperatures above 100°C are reached, enough to cook, but slowly.

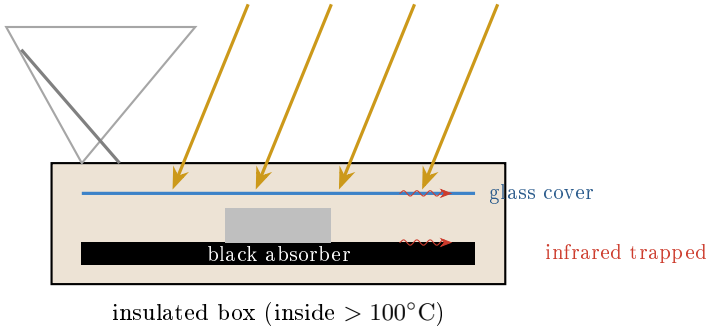


Fig. 3. Solar box cooker cross-section: reflector, glass, black absorber, insulation.

India's largest solar steam kitchen. At the Brahma Kumaris ashram, Mount Abu (Rajasthan), working since the 1990s: 84 parabolic concentrators on the roof heat a receiver and make steam; it can cook for more than 38,000 people.

Daylighting. In the northern hemisphere the south side of a building receives the most sun, so buildings are designed south-facing with large south windows. Passive systems have no operating cost and need no pumps, fans or electricity, but depend on climate and fail under cloud.

4 Active solar thermal: collect, move, store

Definition

An **active** system uses pumps or motors to move heated fluid or air to storage for later use. Components: **solar collector** (on the roof) + **storage tank**. Collectors are classified as **non-concentrating** (area catching light = absorber area) or **concentrating** (catching area > absorber area).

4.1 Flat-plate collector (non-concentrating)

A metal box with a transparent glass cover on top, a black absorber plate, parallel copper pipes carrying water, and insulation on all other faces. It sits on a roof facing south (northern hemisphere) at an inclination of 30–60° to the horizontal. Heated water goes to the tank; uses: domestic hot water, swimming pools in warm climates.

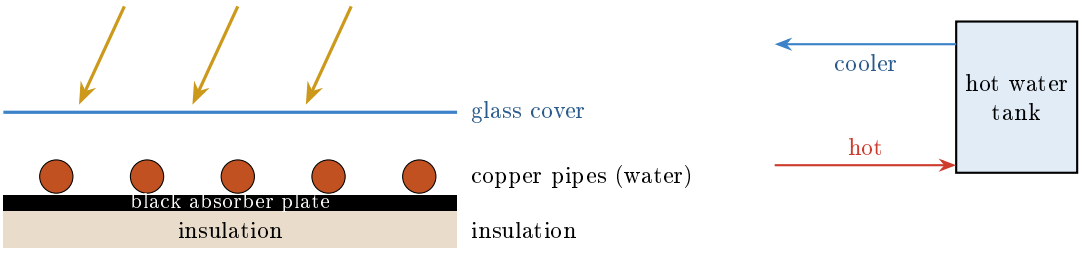


Fig. 4. Left: cross-section of a flat-plate collector. Right: water loop to the storage tank (pump in an active system).

4.2 Concentrating collectors, and why concentration buys temperature

WHY this is true

A concentrator focuses light from aperture area A_a onto a smaller absorber area A_r; the concentration ratio is $C = A_a/A_r$. An ideal black absorber that loses energy only by radiation reaches equilibrium when $C\,S = \sigma T^4$, so

$$T_{\max} = \left(\frac{C\,S}{\sigma}\right)^{1/4} \qquad (S = 1000\ \mathrm{W/m^2}).$$

Because of the fourth root, 10× more concentration gives only $10^{1/4} = 1.78\times$ higher absolute temperature. The limit is the sun's own surface temperature ($\sim$ 5800 K): you cannot heat something hotter than the thing that is heating it.

C	$T_{\max}$ (K)	$T_{\max}$ (°C)	Typical use
1	364	91	flat plate / box cooker (real ones reach > 100°C with glass and insulation)
100	1152	879	trough/dish power plants
1000	2049	1776	central tower
10,000	3644	3371	solar furnace (course: > 3500°C at Mont-Louis, France, operating since 1970)

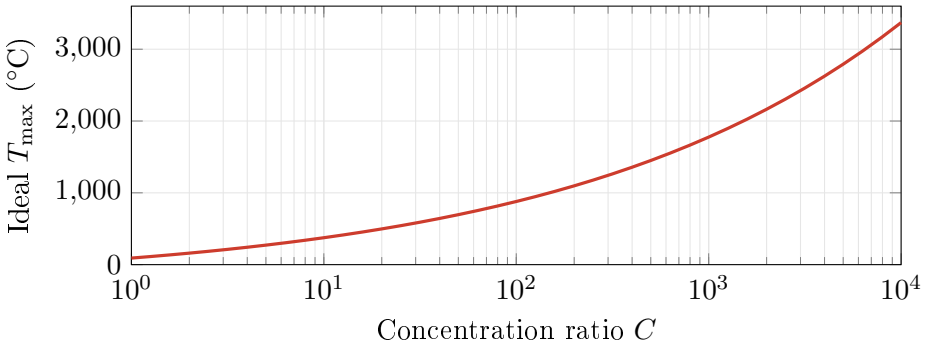


Fig. 5. Temperature gains slow down as C rises.

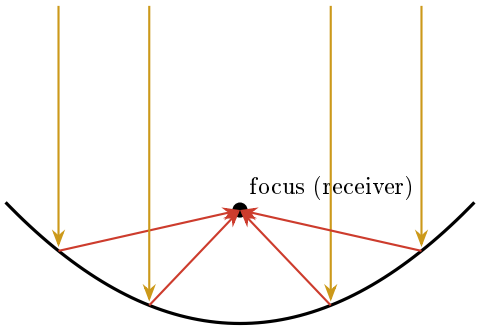


Fig. 6. Why a parabola: $y = x^2/(4f)$ with $f = 1.5$ reflects all rays parallel to the axis through one point.

Solar furnace and heliostats. A solar furnace reflects sunlight from parabolic mirrors or heliostats (sun-tracking flat mirrors) onto a focal point to make very high temperatures.

4.3 Solar thermal electricity and the Carnot limit

WHY this is true

Concentrated heat makes steam that drives a turbine and generator, as in any thermal plant. No heat engine can beat the Carnot efficiency $\eta_{\max} = 1 - T_c/T_h$ (temperatures in kelvin). Hotter input means higher possible efficiency, which is why concentration matters.

Worked example

Solar tower steam at 600°C (873 K), condenser at 30°C (303 K): $\eta_{\max} = 1 - 303/873 = 65\%$ (real plants achieve roughly 35–40%).
Compare OTEC (Week 3 notes): 25°C (298 K) warm water, 4°C (277 K) deep water: $\eta_{\max} = 1 - 277/298 = 7.0\%$, so the observed 1–3% is not bad engineering; it is thermodynamics.

Solar cooling. Same idea as conventional refrigeration (a liquid refrigerant evaporates and absorbs heat from the surroundings), but solar *heat* replaces electricity to drive the cycle.

5 Photovoltaics: turning light directly into electricity

Fixing a common misconception

"Solar panels and solar water heaters are the same thing." A solar water heater is thermal (light → heat). A PV panel is electronic (light → electricity, with no moving parts).
"Photovoltaic means electricity produced by a chemical reaction." The course gloss is misleading: PV involves no net chemical change. "Photo" is light; "voltaic" honours Volta and refers to voltage.
"Hotter sun means more PV power." More *light* increases current, but heat *lowers* voltage and efficiency (see below).

5.1 The physics in four steps

WHY this is true

1. Photon energy. $E = hc/\lambda = 1240\ \text{eV nm}/\lambda$. Silicon has band gap $E_g = 1.12$ eV, so only photons with $\lambda < 1240/1.12 = 1107$ nm can free an electron. Longer-wavelength photons pass through (wasted); shorter ones free an electron but the excess energy becomes heat. This is why even a perfect single-junction silicon cell cannot reach 100% (theoretical ceiling $\sim$ 30–33%).
2. Doping. Pure silicon has 4 valence electrons. Adding **phosphorus** (5) gives spare electrons (n-type); adding **boron** (3) leaves missing electrons called holes (p-type).
3. The junction. Where n meets p, electrons and holes diffuse across and leave fixed ions behind, building an internal electric field that points from n to p.
4. Light. A photon makes an electron–hole pair. The field sweeps electrons to the n side and holes to the p side. Connect the two contacts through a load and electrons flow through it: current.

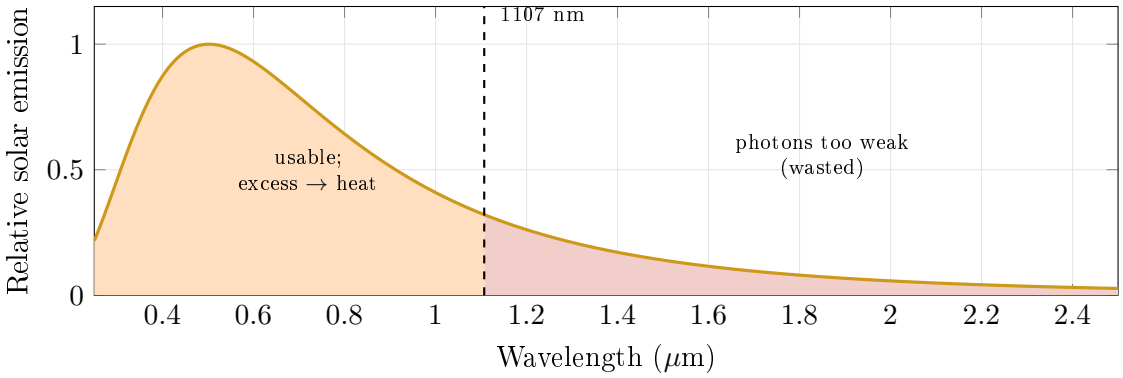


Fig. 7. The sun's spectrum against silicon's cutoff.

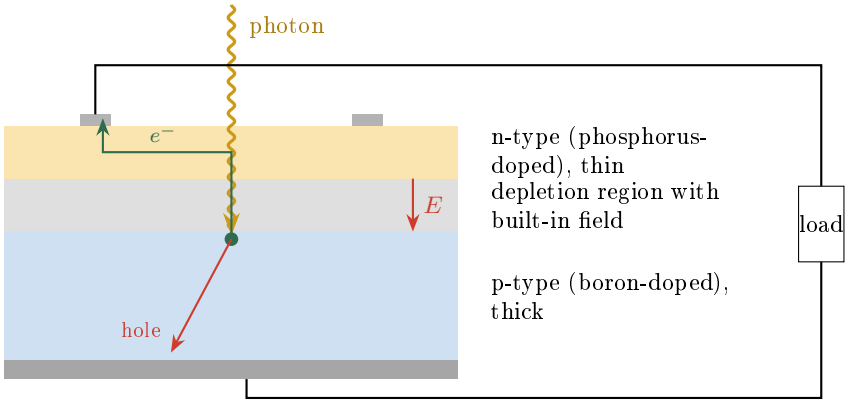


Fig. 8. A silicon p–n junction cell. Electrons leave through the top contact, run through the load, and return at the bottom contact.

5.2 Cell, module, array: why series adds volts and parallel adds amps

One silicon cell gives only $\sim$ 0.6–0.7 V. Joining N_s cells *in series* adds voltages (same current flows through all); joining strings *in parallel* adds currents. A 60-cell module has $V_{oc} \approx 60 \times 0.67 = 40$ V. Modules form panels, panels form arrays, and arrays feed the grid. Output is **DC**, so an **inverter** converts it to AC for appliances and the grid.

5.3 Reading a PV curve and doing the sums

WHY this is true

At $V = 0$ the cell gives its largest current, I_{sc} (short circuit); at $I = 0$ the largest voltage, V_{oc} (open circuit). Power $P = VI$ is zero at both ends and peaks at the **maximum power point** (MPP). **Fill factor** $FF = P_{\max}/(V_{oc}I_{sc})$ measures how "square" the curve is. Efficiency $\eta = P_{\max}/(G\,A)$ with irradiance $G = 1000$ W/m^2.

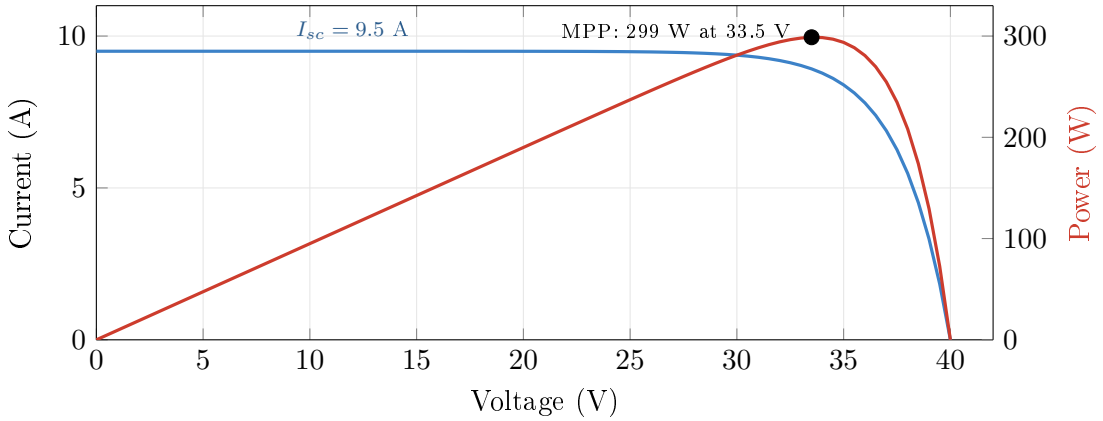


Fig. 9. Model of a 60-cell module ($V_{oc} = 40$ V, $I_{sc} = 9.5$ A, ideal single-diode curve). MPP ≈ 299 W at 33.5 V, 8.9 A.

Worked example

From the model: $FF = 299/(40 \times 9.5) = 0.79$. Module area 1.6 m^2: $\eta = 299/(1000 \times 1.6) = 18.7\%$ (course: commercial panels 10–300 W).
Daily energy: 299 W × 5 peak-sun hours ×0.8 (system losses) ≈ 1.2 kWh/day.
Heat penalty: power falls about 0.4% per °C above 25°C; a cell at 65°C loses 0.004×40 = 16%, giving ≈ 251 W.

Materials and uses (course). Silicon with boron/phosphorus impurities; cadmium telluride, copper indium gallium diselenide; arsenic and antimony have also been used; emerging organic, quantum-dot and hybrid perovskite cells. A cell is thinner than a human hair. Uses: satellites, remote villages and street lighting, desalination and water pumping, telecom devices and railway signals, calculators and watches, and large plants such as Bhadla (14,000 acres, 2250 MW, Rajasthan, the world's largest).

Check against your course notes

The course calls the perovskite example "calcium titanate". *Perovskite* is a crystal structure named after the mineral $CaTiO_3$; solar-cell perovskites are usually lead-halide compounds. For MCQs follow the course wording; for understanding, treat "perovskite" as the structure.

6 Biomass energy

Definition

Biomass is organic material or waste produced by plants or animals: wood, crop residues (wheat/rice straw, groundnut shells, jute sticks, maize cobs), oilseeds, latex plants, starch and sugar crops, municipal and industrial solid waste, sewage, cattle dung. Plant-based biomass is mostly **lignocellulosic** (cellulose, hemicellulose, lignin). Biomass is **indirect solar energy**: photosynthesis stores sunlight as chemical energy. Three product families: **bioenergy** (heat, electricity), **biofuels** (bioethanol, biodiesel) and **biogas**.

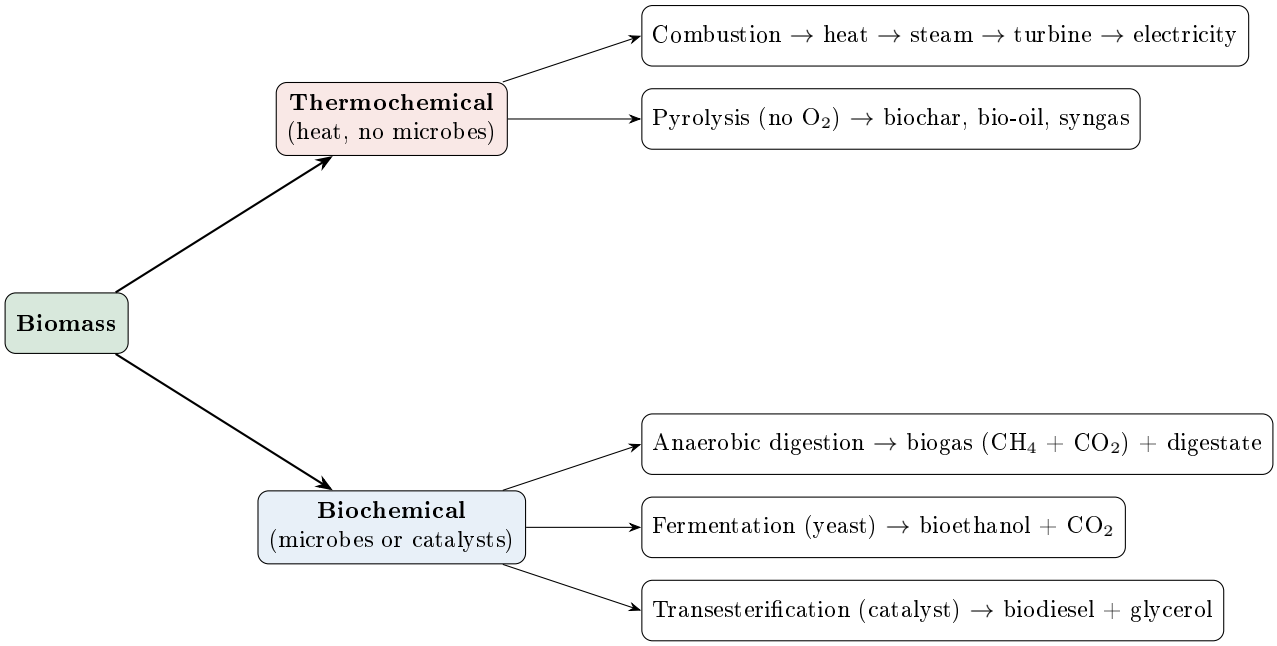


Fig. 10. Conversion routes (the course lists combustion, digestion, pyrolysis, fermentation and catalysed reactions).

6.1 Combustion and the carbon ledger

Burn the biomass, boil water, spin a turbine, drive a generator. Best suited to countries where fuelwood grows fast (tropical countries).

WHY this is true

Burning releases the CO_2 the plant took in while growing, so the *cycle* is roughly carbon-neutral. Check with chemistry: dry wood is about 50% carbon by mass, so 1 t of wood contains 0.5 tC and releases $0.5 \times 44/12 = 1.83$ tCO_2. A fast-growing crop regrows (and re-absorbs this) within months or years; fossil carbon was stored for millions of years, so burning it adds *new* carbon to the fast cycle.

Fixing a common misconception

"Carbon-neutral means zero emissions." The chimney still emits CO_2. Neutrality holds only if the carbon is *re-absorbed within a useful time* and if growing, harvesting and transporting the fuel do not burn much fossil energy. Cutting a slow-growing forest can leave a long "carbon debt". This is why the course also lists deforestation as a risk. (Week 5 BECCS goes further: capture the CO_2 and it becomes carbon-*negative*.)

Worked example

1 t dry wood at ~ 18 GJ/t $= 18{,}000$ MJ $/3.6 = 5000$ kWh of heat. At 25% electrical efficiency: 1250 kWh of electricity. The cooler the steam, the lower the Carnot ceiling, which is why small biomass plants are inefficient.

6.2 Anaerobic digestion and biogas

Definition

Anaerobic digestion is microbial breakdown of biodegradable waste with no oxygen, producing **biogas** (methane about 65%, plus CO_2, hydrogen sulphide, moisture) and **digestate** (a fertiliser-rich residue). A biogas plant is an **anaerobic digester**.

WHY this is true

Why methane and CO_2? Balance atoms for a general organic molecule $C_nH_aO_b$ turning into x CH_4 + y CO_2 using w H_2O:
C: $n = x + y$; H: $a + 2w = 4x$; O: $b + w = 2y$. Solving gives

$$x = \tfrac{n}{2} + \tfrac{a}{8} - \tfrac{b}{4}, \qquad y = \tfrac{n}{2} - \tfrac{a}{8} + \tfrac{b}{4}.$$

Glucose $C_6H_{12}O_6$: $x = 3 + 1.5 - 1.5 = 3$, $y = 3 \Rightarrow 50\%$ CH_4. **A fat** (palmitic acid $C_{16}H_{32}O_2$): $x = 8 + 4 - 0.5 = 11.5$, $y = 4.5 \Rightarrow 72\%$ CH_4. The more hydrogen-rich and oxygen-poor the waste, the more methane, consistent with the course's $\sim 65\%$ for mixed feed.

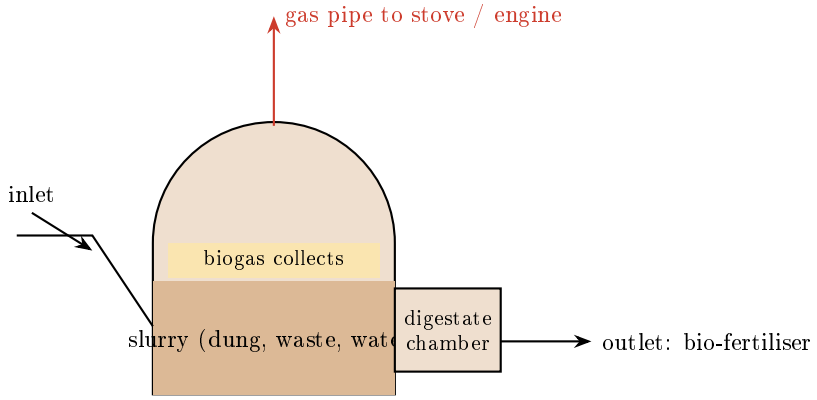


Fig. 11. Dome-type biogas plant. Feed enters the air-tight digester; gas rises to storage; the leftover digestate leaves as fertiliser.

Worked example

Household sizing (rule-of-thumb values). Cooking needs ~ 0.34 m^3 biogas per person per day: 5 people $\Rightarrow 1.7$ m^3/day. Fresh cattle dung gives ~ 0.04 m^3/kg, so $1.7/0.04 \approx 43$ kg dung/day, about 4–5 cattle (India has the world's largest cattle population: course).
Energy: methane ≈ 36 MJ/m^3; at 65% ≈ 23 MJ/m$^3 \approx 6.5$ kWh/m^3.
Climate: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$, so 1 t $CH_4 \rightarrow 44/16 = 2.75$ t CO_2. With $GWP_{100} \approx 28$: vented $= 28$ tCO_2e, burned $= 2.75$ (biogenic) $\Rightarrow \approx 90\%$ cut. Capturing methane is the climate gain.

Check against your course notes

The course says biogas burns "without producing harmful gases such as CO_2, CO, NO_2, SO_2". Strictly, burning biogas *does* release CO_2 (biogenic, from recently fixed carbon) and traces of other gases; what is true is that it burns cleanly, without smoke or soot, and avoids methane escaping. Use the course wording in an MCQ; know the nuance for real understanding. Disadvantages in the notes: inefficient and uneconomical on a large scale; large methane leaks cause a greenhouse effect.

6.3 Pyrolysis: heat without oxygen

Definition

Pyrolysis is thermochemical decomposition of biomass at high temperature in the *absence of oxygen*; the process is irreversible. Main product is a solid, **charcoal/biochar**; industrial pyrolysis can also make liquid fuel and syngas.

WHY this is true

Without oxygen the material cannot burn; heat just drives off water and volatile compounds (rich in oxygen), leaving carbon-rich char. Removing water and oxygen raises energy per kilogram: course says charcoal has about **double** the energy of the original biomass, and it is easier to transport. Industrial conditions in the notes: pressure and above 800°F; convert: $(800 - 32) \times 5/9 = 427°C$.

6.4 Fermentation and bioethanol

Definition

Fermentation: yeast or bacteria convert sugars into alcohol, acids or gases. **Bioethanol** is made from sugary/starchy crops (sugarcane, corn, sorghum). Blends: **E10** = 10% ethanol +90% gasoline; **E85** = 85%/15%. E100 (Brazil hydrous ethanol) and flex-fuel vehicles exist.

WHY this is true

Reaction: $C_6H_{12}O_6 \rightarrow 2\,C_2H_5OH + 2\,CO_2$. Mass: 180 → 92 ethanol + 88 CO_2, so the ideal yield is 92/180 = 0.511 kg ethanol per kg glucose. Starch hydrolyses first, $(C_6H_{10}O_5)_n + nH_2O \rightarrow nC_6H_{12}O_6$, giving 180/162 = 1.11 kg glucose per kg starch.

Worked example

1 t starch → 1.111 t glucose → $1.111 \times 0.511 = 0.568$ t ethanol. At density 0.789 kg/L this is ≈ 720 L (theoretical maximum; real plants reach ∼ 90–95% of it).

Why ethanol lowers fuel economy: energy per litre: ethanol ≈ 21.1 MJ/L, gasoline ≈ 32 MJ/L. E10: $0.1(21.1) + 0.9(32) = 30.9$ MJ/L = 96.6% of gasoline, about 3.4% less energy; the course quotes 2–3% lower fuel economy since higher octane partly compensates. E85: $0.85(21.1) + 0.15(32) = 22.7$ MJ/L = 71% of gasoline.

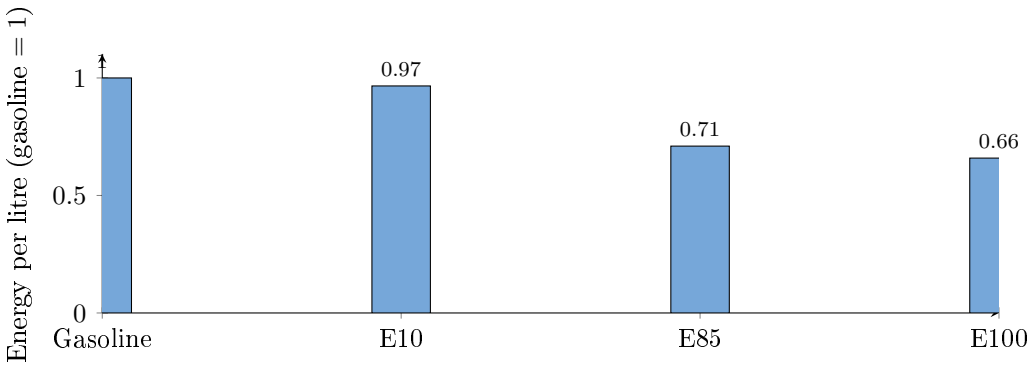


Fig. 12. Volumetric energy of ethanol blends (computed above, using 21.1 and 32 MJ/L).

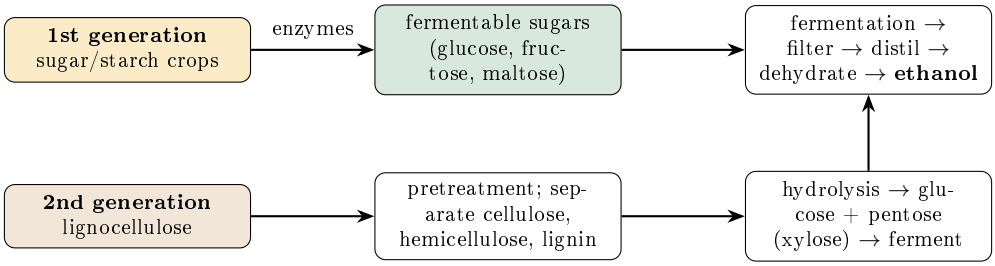


Fig. 13. Why 2nd generation is harder: lignin shields cellulose, and xylose needs special fermentation. Generations (course figure): 1st sugar/starch crops; 2nd lignocellulosic; 3rd algae; 4th synthetic/engineered.

Fixing a common misconception

"Bioethanol runs any car." The course says it *cannot* be used alone in ordinary cars and is blended (E10, E85). Regular cars take low blends (E10–E25); higher blends need **flex-fuel** or dedicated engines (Brazil's E100 shows this works with suitable engines).

Advantages (course): clean fuel, raises octane number, lowers greenhouse emissions; farmer-owned plants support rural economy; a desirable alternative to gasoline. Drawback: lower calorific value, so 2–3% lower fuel economy.

6.5 Biodiesel

Definition

Biodiesel is a renewable, biodegradable liquid fuel made of long-chain fatty-acid *esters* (FAME) from vegetable oils and animal fats plus an alcohol (methanol or ethanol) with a catalyst. It has properties similar to petroleum diesel and can be used alone or blended.

WHY this is true

Transesterification: triglyceride $+\,3\,CH_3OH \xrightarrow{\text{catalyst}} 3$ fatty-acid methyl esters + glycerol.
Why needed: raw oil is too viscous for diesel engines; swapping the glycerol backbone for small methyl groups gives molecules that flow and burn like diesel.
Free fatty acids in the oil would form soap with a basic catalyst, so oils with many of them are first treated with an **acid-catalysed esterification** (the course figure).

Worked example

Triglyceride ≈ 885 g/mol, methanol 32, FAME ≈ 296, glycerol 92. Check: $885+3(32) = 981 = 3(296) + 92 = 980$. From 1000 kg oil: methanol ≈ 108 kg, biodiesel ≈ 905 kg, glycerol ≈ 104 kg.

Sources and India. Rapeseed, soybean, oil palm, sunflower, jatropha, animal fats. India relies mainly on **jatropha**, a shrub of tropical countries (India, Brazil, Africa) that grows in hot, dry areas without frequent rain or fertiliser; it is non-edible, so it avoids competing with food crops. Indian Railways plans large-scale use in locomotives to cut oil bills.

6.6 Biomass versus fossil fuels, and overall pros and cons

	Biomass	Fossil fuels
Renewal time	seasons to decades ("renewable quickly")	millions of years
Carbon	takes carbon from air as it grows; low *net* CO_2	releases ancient carbon: net addition
Supply	continuous (waste never stops)	finite
Cost (course)	cheaper than drilling and pipelines; revenue for farmers	large investment in drilling, extraction, pipelines, distillation

Advantages (course): always available; carbon neutrality; cheaper; extra farmer income; burning solid waste cuts landfill need by 60–90%; fewer greenhouse emissions; reduces fossil dependence (best renewable *liquid* fuels for transport).
Disadvantages: bioethanol and biodiesel are less efficient than fossil fuels; deforestation risk; plants are hard to site near cities because people fear toxic emissions.

7 Hydrogen energy and the solar–hydrogen cycle

Definition

Hydrogen is the simplest element (one proton, one electron). It is abundant but almost always *combined* (water, hydrocarbons), not free as H_2 gas. So hydrogen is an **energy carrier**, not an energy source: it must be made using some primary energy (fossil, nuclear, solar, wind, biomass, hydro, geothermal), stored or moved, then converted to heat or electricity. Burning it, or reacting it in a fuel cell, produces water vapour and no CO_2.

Fixing a common misconception

"Hydrogen is a source like oil." There are no hydrogen wells; making H_2 costs energy (the course's "net energy is negative" point). It is like a battery or electricity: valuable for *moving and storing* energy.

"Hydrogen is energy-dense." Per *kilogram*: yes, the highest of any common fuel (~ 120 MJ/kg versus ~ 44 for gasoline, 2.7×). Per *litre*: no, because it is so light. Even at 700 bar (about 40 kg/m^3): $40 \times 120/1000 = 4.8$ MJ/L versus gasoline 32 MJ/L, only 15%. Per litre at 1 atm (0.0899 kg/m^3) it is ~ 0.01 MJ/L. This is why storage is the main vehicle problem, yet it suits aircraft, where mass matters more than volume.



Fig. 14. Same fuel, opposite verdicts depending on whether you count mass or volume.

7.1 Five production routes

Route	Core idea	Climate profile
Natural-gas (steam) reforming	hot steam reacts with a hydrocarbon	fossil; releases CO_2
Thermochemical water splitting	500–2000°C heat (concentrated solar or nuclear waste heat) drives chemical cycles	near-zero GHG if the heat is solar/nuclear
Electrolysis	electricity splits water	depends on the electricity source
Solar-driven	light as the agent (photobiological, photoelectrochemical, solar thermochemical)	near-zero GHG
Biological	microbes (bacteria, microalgae) from biomass or wastewater	renewable, early stage

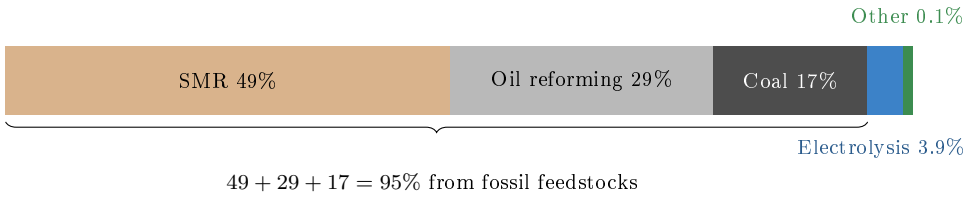


Fig. 15. Course pie chart of world hydrogen production.

Check against your course notes

The text says "> 95% of hydrogen is produced from steam reforming of natural gas", while the pie gives SMR 49%. The two agree if you read the text loosely: fossil feedstocks together ($49 + 29 + 17 = 95\%$) supply about 95%. Learn both: $> 95\%$ and SMR 49%, oil 29%, coal 17%, electrolysis 3.9%, other 0.1%.

7.2 Steam methane reforming (SMR): a mass balance

WHY this is true

Step 1 (endothermic, steam reforming): $CH_4 + H_2O \rightarrow CO + 3H_2$. Step 2 (water-gas shift): $CO + H_2O \rightarrow CO_2 + H_2$. Net: $CH_4 + 2H_2O \rightarrow CO_2 + 4H_2$.
Masses: $16+36 = 52 \rightarrow 44+8$. So per tonne of H_2: 2 t CH_4, 4.5 t water, 5.5 t CO_2. The course diagram uses 1.1 t H_2: CH_4 2.2 t, water $\approx$ 4.9 t, $CO_2 \approx$ 6.0 t. These match the stoichiometry. With carbon capture ($\sim$ 70% in the diagram) only about 1.5 t escapes. Disadvantages (course): natural gas is becoming scarce and costly, and the process releases CO_2.

7.3 Electrolysis: where the energy numbers come from

WHY this is true

Splitting one mole of liquid water, $H_2O \rightarrow H_2 + \frac{1}{2}O_2$, needs $\Delta H = 285.8$ kJ in total, of which $\Delta G = 237.1$ kJ must be supplied as electrical work (the rest, $T\Delta S = 48.7$ kJ, can come as heat). Electrical work per mole is nFE with $n = 2$ electrons:

$$E^\circ = \frac{\Delta G}{nF} = \frac{237{,}100}{2 \times 96{,}485} = 1.23 \text{ V}, \qquad E_{\text{thermoneutral}} = \frac{\Delta H}{nF} = 1.48 \text{ V}.$$

Per kilogram: 285.8/2.016 = 141.8 MJ = 39.4 kWh (the theoretical energy content, HHV). Real electrolysers need about 50–55 kWh/kg ($\sim$ 70–75% efficient).

Worked example

Mass balance of $2H_2O \rightarrow 2H_2 + O_2$: $36 \rightarrow 4 + 32$, so 9 t water gives 1 t H_2 and 8 t O_2 (oxygen is a useful by-product). Reaction heat is why high-temperature electrolysis or thermochemical cycles can cut the electricity needed: heat supplies the $T\Delta S$ share.

Check against your course notes

The course electrolysis diagram labels 1.1 t H_2 with 8 t O_2 and 9.1 t water. Total mass balances ($1.1 + 8 = 9.1$), but stoichiometry fixes O_2:H_2 at 8:1 by mass, so the labels are a rounded illustration. Reactions to memorise: *anode* $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$; *cathode* $4H^+ + 4e^- \rightarrow 2H_2$; PEM electrolysis runs at 20–100°C.

7.4 Thermochemical, solar-driven and biological routes

Thermochemical cycles use reusable chemicals (cerium oxide, copper chloride) so only water is consumed; direct thermal splitting of water needs impractically high temperatures, so the cycle splits the job into steps: a very hot step releases oxygen from the oxide, a cooler step lets the oxide take oxygen from steam, freeing H_2. **Solar-driven** processes: *photobiological* (bacteria and green algae use photosynthetic activity), *photoelectrochemical* (specialised semiconductors split water directly; the semiconductor must deliver at least 1.23 V, so a band gap of roughly 1.6–2 eV in practice, which is why plain silicon at 1.12 eV is not enough alone), and *solar thermochemical* (concentrated solar heat with metal oxides). **Biological** routes use microbes breaking down biomass or wastewater.

7.5 Fuel cells, and a classic naming trap

Definition

A **fuel cell** converts chemical energy directly into electrical energy (like a battery, but fed continuously with fuel). Hydrogen goes to the **anode**, where it splits into protons and electrons. Electrons travel through the external circuit (current); protons cross the membrane; at the **cathode** they meet oxygen and electrons to form water. Outputs: **electricity, heat, water**.

Fixing a common misconception

Anode is defined by *oxidation* ("An Ox"), not by sign. In the electrolyser the anode (water oxidised to O_2) is the *positive* electrode; in the fuel cell the anode (H_2 oxidised) is the *negative* electrode. The fuel cell is electrolysis run backwards.

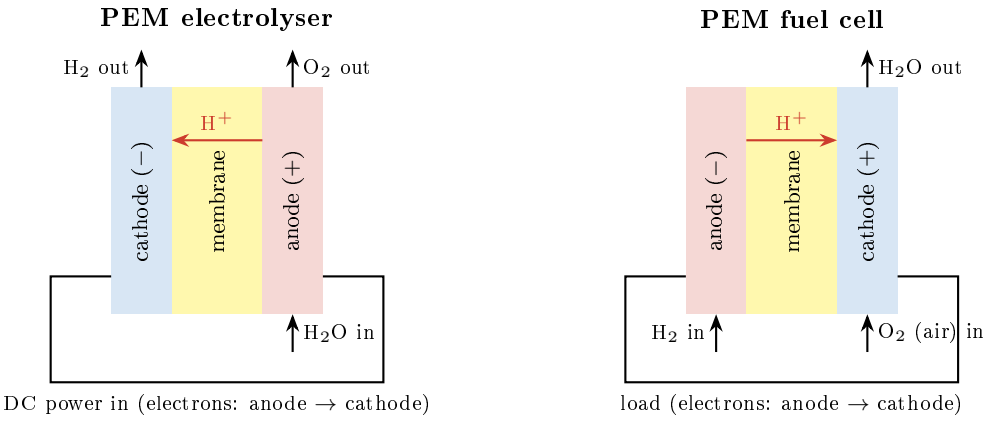


Fig. 16. Same layers, reversed flows. In both, oxidation (the anode reaction) is where electrons are released.

Worked example

Real cell voltage under load ≈ 0.7 V (ideal 1.23 V; losses from reaction kinetics, resistance, gas transport). Efficiency on the HHV basis $= 0.7/1.48 = 47\%$, versus roughly 20–25% for a petrol engine: about 2× (course: two to three times). A fuel cell is not limited by Carnot because it does not run on heat.

7.6 The solar–hydrogen cycle ("solar-hydrogen revolution")

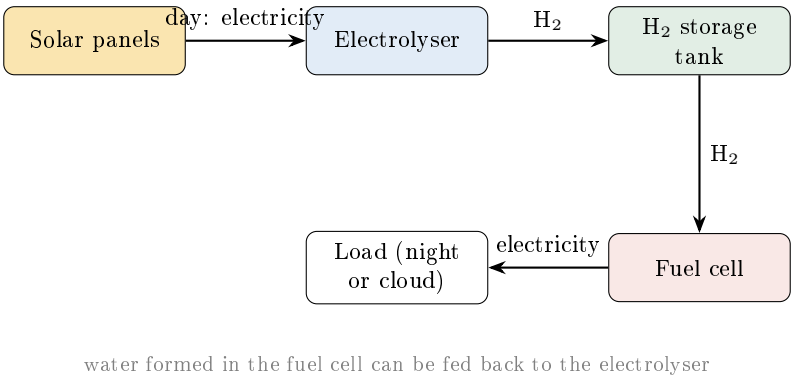


Fig. 17. Three components (course): electrolyser, storage tank, fuel cell. By day, surplus solar power makes hydrogen; at night, the fuel cell runs the load.

Worked example

A 5 kW load for 12 night hours needs 60 kWh. Fuel cell at 50% (HHV) $\Rightarrow$ 120 kWh of hydrogen energy $= 120/39.4 = 3.05$ kg H_2. Making it at 70% electrolyser efficiency takes $120/0.7 = 171$ kWh of daytime solar electricity. **Round trip** $= 60/171 = 35\%$. Storage: at 350 bar ($\sim$ 24 kg/m^3) 3.05 kg takes 0.13 m^3 (127 L), but at atmospheric pressure it would fill $\sim$ 34 m^3. So a battery ($\sim$ 90% round trip) wins for daily cycling, while hydrogen wins for long-duration or seasonal storage and heavy transport.

7.7 Hydrogen economy, uses, pros and cons

The **hydrogen economy** is the vision of hydrogen as a low- or zero-carbon energy carrier: made from diverse sources, stored, transported, and converted to heat and electricity (replacing natural gas for heating and petrol as transport fuel). Uses in the course: oil refining and fertilisers (ammonia, $N_2 + 3H_2 \rightarrow 2NH_3$) today; vehicles and buses; home fuel-cell units (about the size of a refrigerator, giving heat, hot water and electricity); Honda's home unit that makes hydrogen from methane, used by many Japanese households; aviation (most energy per gram). Small fuel cells power laptops and phones; large ones supply grids, emergency power and off-grid sites.

Advantages: plentiful water feedstock, low environmental impact, renewable if made from renewables (solar-hydrogen), good oil substitute, easier to store than electricity, safer than gasoline and natural gas (course), efficient fuel cells now commercial.

Disadvantages: not found as H_2, so energy must be spent to make it (net energy negative); hard to store in vehicle tanks; fuel cells are currently expensive.

8 Quick recall with derivations: hydro, wind, ocean thermal, geothermal

Full illustrated notes for these four are in your dashboard Week 3; this section adds the numbers and reasoning that make the facts stick.

8.1 Hydroelectric power

WHY this is true

Water of mass m at height H stores mgH. Per second, $\dot{m} = \rho Q$, so the electrical power is $P = \eta\,\rho\,g\,Q\,H$ (η efficiency, Q flow m^3/s). Chain: gravitational potential → kinetic (in the penstock) → mechanical (turbine, shaft) → electrical (generator). Hydropower efficiency exceeds 80% whereas thermal plants are as low as 40% (course) because no heat-to-work conversion is involved, so no Carnot limit.

Worked example

$Q = 100\ \mathrm{m}^3/\mathrm{s}$, $H = 50$ m, $\eta = 0.9$: $P = 0.9 \times 1000 \times 9.81 \times 100 \times 50 = 44$ MW.
Pumped storage: energy in $V = 10^6\ \mathrm{m}^3$ raised 300 m: $E = \rho g H V = 1000 \times 9.81 \times 300 \times 10^6 = 2.9 \times 10^{12}$ J $= 817$ MWh. Surplus off-peak power pumps water up; peak demand releases it. Dam + two reservoirs at different heights is the key difference from a plain dam; run-of-river uses natural flow at small scale.

Course facts: $>$ 18% of world electricity, fourth-largest source; Romans used waterwheels for grinding; first hydroelectric plant: Fox River, Wisconsin, 30 Sept 1882; plants last at least 40–50 years. Drawbacks: expensive dams, altered landscape, habitat loss, displacement of people (especially tribal communities), flooding of large areas, disputes between states or countries.

8.2 Wind energy

WHY this is true

Moving air of density ρ through rotor area A at speed v carries kinetic power $\frac{1}{2}\rho A v^3$. A turbine captures a fraction C_p, and no turbine can exceed the **Betz limit** $16/27 = 59\%$ (the wind must keep moving to leave the rotor). Because of v^3, a site with twice the wind gives eight times the power: siting matters more than almost anything else.

Worked example

Rotor diameter 100 m ($A = 7854\ \mathrm{m}^2$), $v = 12$ m/s, $\rho = 1.225\ \mathrm{kg/m}^3$: available $= \frac{1}{2}(1.225)(7854)(12^3) = 8.3$ MW; with $C_p = 0.40$: 3.3 MW. At 6 m/s: 0.4 MW for the same C_p (one-eighth).

Course facts: wind is indirect solar (uneven heating → pressure differences). Ancient Egyptians sailed with wind $\sim$ 4000–3500 BC; windmills appeared in the 8th–9th centuries (Middle East/Western Asia). Horizontal-axis turbines are most common and more efficient; vertical-axis have a vertical shaft. Gearbox → generator → converter → grid. Wind farms feed the grid more easily. Tamil Nadu and Gujarat lead India (NIWE); India is fifth in the world. Drawbacks: unreliable wind, noise, remote sites, visual pollution, radio/TV interference, bird and bat deaths, costly setup.

8.3 Ocean thermal energy (OTEC)

WHY this is true

Needs a temperature difference $> 20°C$ between warm surface and cold deep water, largest (20–25°C) in the tropics. The Carnot ceiling for 25°C and 4°C is 7% (Section 4), explaining the real 1–3%. **Closed cycle:** warm water boils a low-boiling working fluid (ammonia −33°C, propane −42°C, R-134a −26.3°C) in a heat exchanger; vapour drives the turbine; cold deep water condenses it; the fluid recycles. **Open cycle:** warm seawater itself is flash-boiled in a low-pressure chamber. Why that works: water boils at 25°C when its saturation pressure is only ≈ 31.6 hPa ($\approx 3\%$ of atmospheric), so a partial vacuum makes warm seawater boil; the steam condenses to **desalinated water**.

Advantages: renewable, clean, reliable (as long as the sun shines), weather-independent, low maintenance, far from settlements. Disadvantages: costly installation, harm to marine life from pumping, low efficiency, location-specific, cost of transmitting power from floating plants to land.

8.4 Geothermal energy

WHY this is true

Temperature rises about 1°C per 35–40 m of depth. With a 15°C surface, drilling the course's "1 or 2 miles" (1.6–3.2 km) gives only $15 + 1600/37 = 58°C$ to $15 + 3200/37 \approx 100°C$ in ordinary rock: warm, not steam-hot. Power plants therefore sit where the gradient is much steeper (volcanic regions, hot springs, geysers, magma "hot spots" where water meets hot rock), which is why geothermal is **location-specific**. Hydrothermal cycle: cold water injected down one well, heated, rises as hot water/steam through another (pressure drops → water flashes to steam), drives a turbine, is condensed in cooling towers and *re-injected*.

Check against your course notes

The course explains Earth's core heat as "nuclear fusion reactions". Fusion does not occur in Earth's interior; the heat comes from leftover formation heat and radioactive decay. Answer in the course's words for MCQs but know the real source. Other course facts: not indirect solar; magma, hot spots, geysers; available for 2–3 billion years; drawbacks include huge up-front cost, earthquake risk, release of H_2S and SO_2.

9 Quick recall: environmental law and EIA (Week 4 in the dashboard)

Law	Year	Key facts from the course
Stockholm Conference	1972	UN Conference on the Human Environment; **5 June** later observed as World Environment Day
Wildlife Protection Act	1971 (course)	sanctuaries and national parks, zoos, trade in wild animals; amended 1982, 1986, 1991, 1993, 2002, 2006; Wild Life Advisory Board, Chief Wild Life Warden; 104 national parks, 551 sanctuaries, 86 conservation reserves, 11 biosphere reserves
Water Act	1974	SPCB consent to establish and to operate; appellate authority; penalties; Central and State Boards
Forest (Conservation) Act	1980	protects forests from illegal cutting, encroachment, fire, grazing, shifting agriculture; reserve forest cannot be de-reserved without Central Government permission
Air Act	1981 (amended 1987)	50 sections; noise pollution added in 1987; consent decided within 4 months; limits on particulates, CO, SO_2, NO_x, VOCs
Environment (Protection) Act	1986	Article 253; in force 19 Nov 1986; 26 sections, 4 chapters; umbrella legislation after the Bhopal gas tragedy (MIC, Union Carbide, 3–4 Dec 1984); standards for 61 industry categories; annual environmental audit report by 30 September

Check against your course notes

The course gives the Wildlife Protection Act as 1971. Many sources date it 1972. Use the course's year in an MCQ.

EIA steps: screening (Schedule 1: always assess, e.g. thermal plants $\geq$ 300 MW; Schedule 2: if significant effects) $\rightarrow$ scoping $\rightarrow$ impact identification/prediction $\rightarrow$ assessment and mitigation $\rightarrow$ EIA report (Environmental Impact Statement) $\rightarrow$ decision-making $\rightarrow$ monitoring and evaluation. **Methods:** ad hoc (opinion poll, expert opinion, Delphi); checklist (simple, descriptive, scaling, weighing); matrix (two-dimensional, Leopold); network (primary and secondary impacts); overlay (transparent maps, GIS).

10 Week 6 synthesis

Resource	Origin	Core physics	Main limitation
Solar thermal	direct sun	absorb, concentrate; $T_{\max} = (CS/\sigma)^{1/4}$	weather, storage
Solar PV	direct sun	band gap, p–n junction	intermittent; DC$\rightarrow$AC; heat lowers output
Hydro	indirect solar (water cycle)	$P = \eta\rho g Q H$	dams: ecology, displacement
Wind	indirect solar	$\frac{1}{2}\rho A v^3 C_p$, Betz 59%	variable; noise; siting
Biomass	indirect solar	photosynthesis, chemistry	land, emissions, low efficiency liquids
OTEC	solar-heated ocean	Carnot, $\Delta T > 20$°C	1–3% efficiency; cost
Geothermal	Earth's interior heat	gradient ~ 1°C/37 m	geology-specific
Hydrogen	carrier (any source)	$\Delta G = 237$ kJ/mol, fuel cell	storage, cost, production energy

Self-test checklist

- ☐ Define net energy and use it to explain the coal–nuclear contrast in the notes.
- ☐ Show that an hour of sunlight is of the order of a year of world energy use.
- ☐ Explain why a solar cooker works in greenhouse-effect terms; give three passive and three active examples.
- ☐ Derive $T_{\max} = (CS/\sigma)^{1/4}$ and explain why 10× concentration gives only 1.78× temperature.
- ☐ Explain doping (P vs B), the junction field and the 1107 nm silicon cutoff; compute FF and efficiency.
- ☐ List the five biomass conversion routes and give one product for each.
- ☐ Derive CH_4/CO_2 ratio for glucose using atom balance; compute 1 t starch → ethanol.
- ☐ Compute energy per litre of E10 and E85 and explain the 2–3% fuel-economy loss.
- ☐ Write the transesterification reaction and explain why free fatty acids are a problem.
- ☐ Explain "energy carrier" and compute the electrolysis voltage $\Delta G/nF$.
- ☐ Explain why hydrogen is energy-dense per kg but not per litre.
- ☐ Distinguish anode and cathode polarity in an electrolyser versus a fuel cell.
- ☐ Do the solar–hydrogen night-load sums and state the round-trip efficiency.
- ☐ State $P = \eta\rho g Q H$, $P = \frac{1}{2}\rho A v^3 C_p$ and the Carnot limit with one example each.